



AI & Partners

Amsterdam - London - Singapore

EU AI Act

Slovenia

AI Ecosystem Brief

December 2025

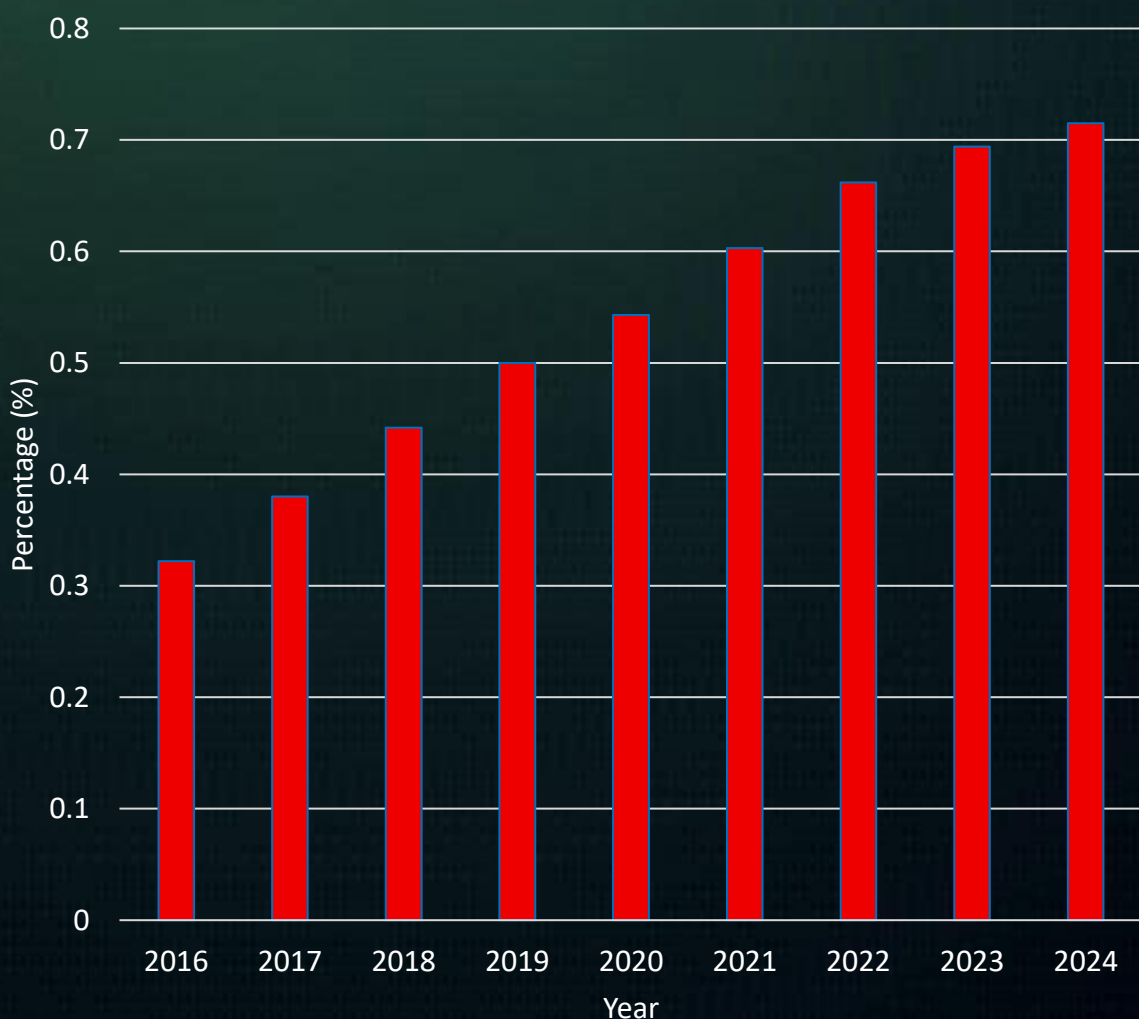
Acknowledgements

Sean Musch, Michael Borrelli, Charles Kerrigan, Kristina Kočet Hudrap, Sara Kremsar, Milovan Tomašević, Aljaz (AI) Peklaj, DAVID HORVAT, Matija Jamnik, Klemen Selakovič, Sasa Sodja, Andrej P. Skraba

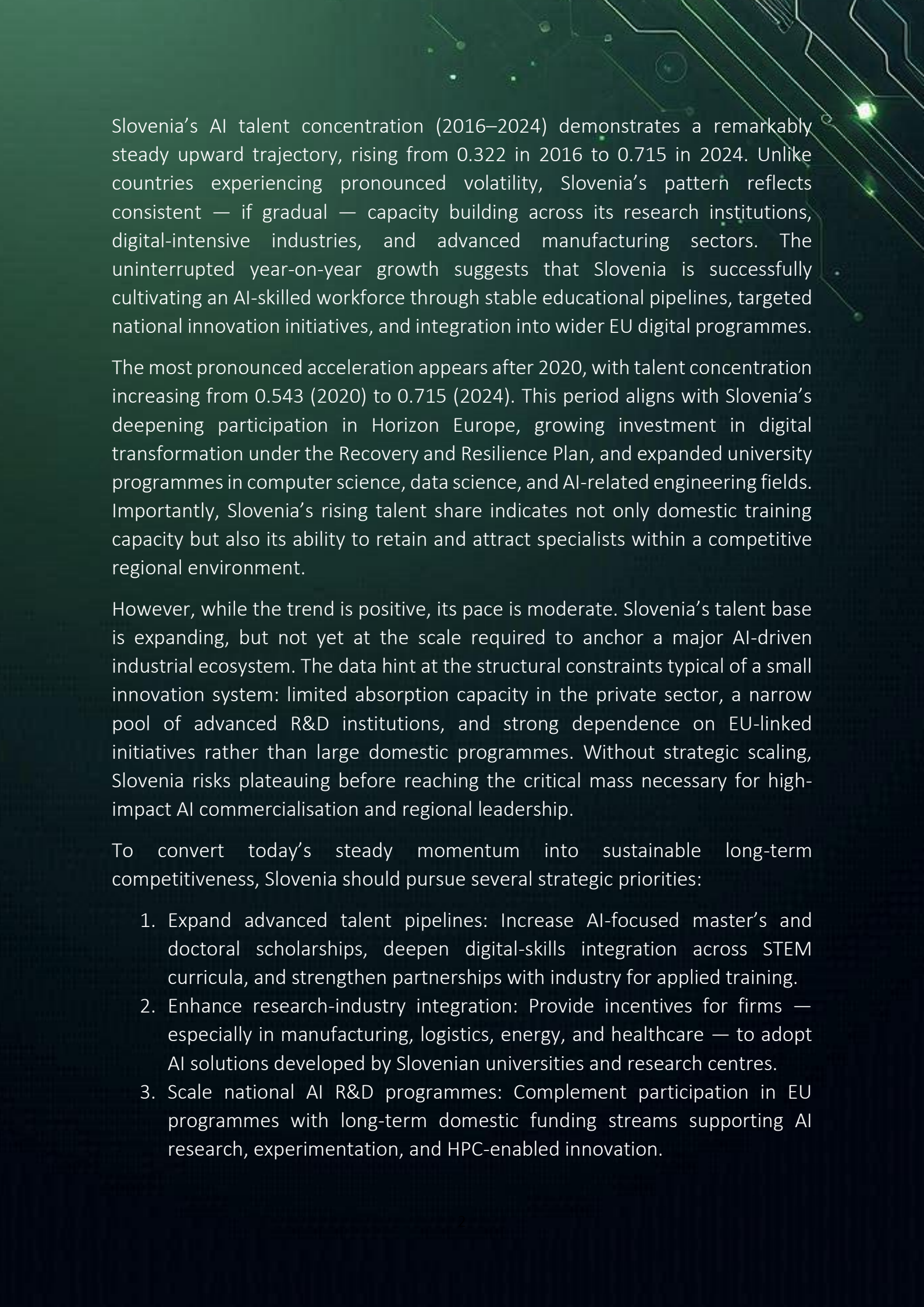


Sustaining Slovenia's AI Talent Momentum: Strategic Priorities to Convert Steady Gains into a Robust, Competitive, and Innovation- Driven Ecosystem by 2030

*AI talent concentration in Slovenia
(November 2025)*



Explanation: This chart shows the concentration of LinkedIn members with at least two AI engineering skills or who perform an AI occupation in Slovenia and in time. The year over year computations start in 2017.



Slovenia's AI talent concentration (2016–2024) demonstrates a remarkably steady upward trajectory, rising from 0.322 in 2016 to 0.715 in 2024. Unlike countries experiencing pronounced volatility, Slovenia's pattern reflects consistent — if gradual — capacity building across its research institutions, digital-intensive industries, and advanced manufacturing sectors. The uninterrupted year-on-year growth suggests that Slovenia is successfully cultivating an AI-skilled workforce through stable educational pipelines, targeted national innovation initiatives, and integration into wider EU digital programmes.

The most pronounced acceleration appears after 2020, with talent concentration increasing from 0.543 (2020) to 0.715 (2024). This period aligns with Slovenia's deepening participation in Horizon Europe, growing investment in digital transformation under the Recovery and Resilience Plan, and expanded university programmes in computer science, data science, and AI-related engineering fields. Importantly, Slovenia's rising talent share indicates not only domestic training capacity but also its ability to retain and attract specialists within a competitive regional environment.

However, while the trend is positive, its pace is moderate. Slovenia's talent base is expanding, but not yet at the scale required to anchor a major AI-driven industrial ecosystem. The data hint at the structural constraints typical of a small innovation system: limited absorption capacity in the private sector, a narrow pool of advanced R&D institutions, and strong dependence on EU-linked initiatives rather than large domestic programmes. Without strategic scaling, Slovenia risks plateauing before reaching the critical mass necessary for high-impact AI commercialisation and regional leadership.

To convert today's steady momentum into sustainable long-term competitiveness, Slovenia should pursue several strategic priorities:

1. Expand advanced talent pipelines: Increase AI-focused master's and doctoral scholarships, deepen digital-skills integration across STEM curricula, and strengthen partnerships with industry for applied training.
2. Enhance research-industry integration: Provide incentives for firms — especially in manufacturing, logistics, energy, and healthcare — to adopt AI solutions developed by Slovenian universities and research centres.
3. Scale national AI R&D programmes: Complement participation in EU programmes with long-term domestic funding streams supporting AI research, experimentation, and HPC-enabled innovation.

- 
4. Increase attractiveness for international experts: Promote Slovenia's quality of life, bolster competitive research contracts, and leverage regional AI initiatives to draw experienced researchers and engineers.
 5. Strengthen monitoring of AI talent dynamics: Establish an annual national AI talent index that tracks graduation rates, workforce flows, industrial demand, and participation in international projects.

In summary, Slovenia's AI talent concentration reflects a stable, consistently improving trajectory that provides a strong basis for future growth. To transform this gradual rise into a resilient, innovation-rich AI ecosystem, Slovenia must scale its talent development, deepen industry engagement, and secure sustained investment. These actions will be essential for positioning Slovenia as a dynamic and competitive contributor to Europe's AI landscape by 2030.



AI & Partners

Amsterdam - London - Singapore

Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

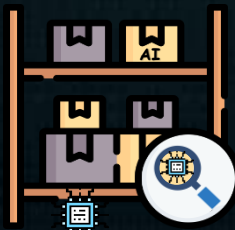
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

Contact: Team | contact@ai-and-partners.com

Contents

This section outlines the structure of the report, detailing chapters on Slovenia's AI policy landscape, regulatory initiatives, governance mechanisms, findings, and recommendations. It serves as a navigation guide, enabling readers to quickly locate specific analyses, datasets, and policy discussions relevant to Slovenia's evolving artificial intelligence ecosystem.

Who Are We	4
Contents	5
Overview	7
Specific AI Governance or Law	9
Specific AI Governance in Slovenia	9
AI Ethics Guidelines and Sectoral Governance Tools	10
Proposed AI Regulatory Authority and Future Outlook	10
Slovenia's National	11
AI Strategy	11
Approach to	13
AI Regulation	13

Approach to Foster Innovation.....	16
Digital Decade 2025: Report.....	19
Government AI Readiness	21
Conclusion.....	24
Appendix A: Expert Vignettes.....	26
Andrej P. Skraba, <i>Co-Founder, Astra AI</i>	27
Biography	27
Kristina Kočet Hudrap, <i>CEO, TIKO PRO</i>	28
Biography	28
Sara Kremsar, <i>Founder, Optimod</i>	29
Appendix B: OECD AI Progress Report.....	30
References.....	35

Overview

Artificial Intelligence (AI) is reshaping economies and governance worldwide, and Slovenia is positioning itself as a regional leader in responsible, EU-aligned AI development.

Grounded in a long tradition of AI research and supported by comprehensive national strategies, Slovenia's governance model emphasizes ethical safeguards, multistakeholder coordination, and legal alignment with the EU AI Act. Rather than introducing sweeping new laws in isolation, Slovenia is leveraging strategic instruments, regulatory infrastructure, and international engagement to build a robust AI ecosystem.

At the heart of Slovenia's AI governance strategy is the National Programme for the Promotion of the Development and Use of Artificial Intelligence by 2025 (NpUI)—a cross-sectoral blueprint adopted in 2021 that sets out ten strategic objectives for AI integration across research, education, public services, and the economy. Additionally, there is also a National programme to promote the development and use of AI by 2025 that covers the years up to 2030, with comments to the draft being submitted up to 18 December 2025. In preparation for the EU AI Act's full application in 2026, Slovenia has designated the Ministry of Digital Transformation as one authority for implementation, supported by an expert council to ensure compliance and multistakeholder input. Based on the newly adopted AI law, that entered into force at end of November 2025, competences are split across multiple authorities. A regulatory sandbox—



targeted for launch by August 2026—will offer a safe space to test high-risk AI systems while ensuring legal and ethical accountability.

Slovenia's AI governance architecture is embedded in a broader digital transformation agenda. In addition to the Ministry of Digital Transformation, national coordination efforts involve universities, public research institutes, private-sector actors, and civil society. The country's Digital Innovation Hub and Strategic Research and Innovation Partnerships foster experimentation and cross-sector collaboration, while awareness campaigns and AI training programs help build institutional and public capacity.

On the global stage, Slovenia is an active participant in shaping ethical AI norms. It hosts the International Research Centre for Artificial Intelligence (IRCAI) under the auspices of UNESCO, contributes to the OECD AI Observatory, and aligns closely with UNESCO's Recommendation on the Ethics of AI. Slovenia is also engaged in EU initiatives such as the Coordinated Plan on AI and the European AI Factories network, through its Slovenian AI Factory (SLAIF)—reinforcing its commitment to interoperable, human-centric AI governance.

Sectorally, Slovenia is deploying AI in domains such as public administration, education, agriculture, healthcare, and environmental protection. The NpUI emphasizes the use of scalable “reference AI solutions” to modernize public services, enhance service delivery, and strengthen trust in government. AI education and workforce reskilling are key pillars of the national strategy, with a focus on university programs, lifelong learning, and equitable access to digital skills.

Slovenia's AI infrastructure strategy is equally ambitious. Investments include the HPC RIVR supercomputing project, participation in EuroHPC, and the development of national data platforms aligned with EU FAIR data principles. The NpUI supports trusted data use, secure AI deployment, and technical interoperability to ensure ethical and scalable innovation.

Through its value-driven, EU-integrated approach, Slovenia is building an adaptive AI governance model rooted in transparency, inclusion, and societal benefit. By harmonizing national efforts with European frameworks and global norms, Slovenia is charting a course for AI that is not only innovative, but also democratic, ethical, and internationally respected.

Specific AI Governance or Law

Slovenia's 2021 AI strategy guides research, innovation, education, and public sector AI use. Aligning with the August 2024 EU AI Act, Slovenia drafted implementing legislation with public consultation in early 2025. The Ministry of Digital Transformation leads enforcement, supported by an expert council and a planned 2026 AI regulatory sandbox.

Specific AI Governance in Slovenia

Slovenia has enacted a dedicated national strategy to guide its AI development—the National Programme for the Promotion of the Development and Use of Artificial Intelligence by 2025 (NpUI), adopted on 27 May 2021. This strategic document outlines ten core objectives across key implementation areas, aiming to build a robust AI ecosystem that supports research, innovation, education, and public sector integration. In alignment with the EU AI Act, which entered into force in August 2024, Slovenia has drafted national legislation to transpose the Act into domestic law. A public consultation on this draft took place in early 2025. The Ministry of Digital Transformation has been designated as the central authority responsible for the Act's implementation. To support this transition, Slovenia has also established an expert council to advise on regulatory and policy issues. By 2026, Slovenia plans to operationalize a national AI regulatory sandbox—an innovation-friendly environment that enables the safe testing of AI systems while ensuring compliance with legal and ethical standards.

AI Ethics Guidelines and Sectoral Governance Tools

While the NpUI serves as the overarching strategic framework, Slovenia has also embedded ethical considerations across sector-specific applications. The programme emphasizes OECD-aligned principles such as human-centric AI, transparency, and public trust.

Public administration is a key focus area. Slovenia is introducing AI solutions to improve governmental services and has launched training initiatives to enhance digital competencies among civil servants. Additionally, the government is working to ensure open access to public sector data, promote citizen awareness, and establish a Digital Innovation Hub to support AI experimentation and knowledge sharing. Slovenia is also promoting ethical AI development internationally through its leadership of the UNESCO-affiliated International Research Centre for Artificial Intelligence (IRCAI), which focuses on the intersection of AI and sustainable development.

While formal legal precedent regarding AI systems remains limited, Slovenia has emphasized the need for human oversight and accountability in AI use, particularly in sensitive sectors. The government's focus on inclusive governance and rights-based AI aligns with emerging European norms, even as sector-specific regulatory enforcement mechanisms are still evolving.

The designation of the Ministry of Digital Transformation as the central regulatory body under the EU AI Act, supported by expert advisory mechanisms, marks a clear step toward structured oversight. It is anticipated that additional institutions will join this regulatory architecture as implementation progresses.

Proposed AI Regulatory Authority and Future Outlook

Slovenia's AI oversight model is grounded in proactive policy development and EU harmonization. With the NpUI providing strategic direction and the Ministry of Digital Transformation leading regulatory coordination, Slovenia is positioned to become a regional leader in ethical AI governance.

As the full application of the EU AI Act approaches in 2026, Slovenia's continued focus on infrastructure development, international cooperation, and human capital investment will be critical. The planned AI regulatory sandbox and public-private research collaborations signal a future-oriented approach to ensuring AI technologies are trustworthy, inclusive, and aligned with societal values.

Slovenia's National AI Strategy

Slovenia's AI governance approach is grounded in a value-driven, EU-aligned framework that combines ethical safeguards, strategic public-private cooperation, and a commitment to digital innovation to foster inclusive, trustworthy, and human-centric AI development. Through instruments like the National AI Programme (NpUI), alignment with the EU AI Act, and multistakeholder coordination, Slovenia promotes a comprehensive and forward-looking model for responsible AI deployment.

Slovenia's AI governance framework is underpinned by core values of inclusivity, human dignity, public trust, and technological sovereignty. These principles are reflected in the National Programme for the Promotion of the Development and Use of Artificial Intelligence by 2025 (NpUI)—a cross-sectoral strategy that outlines 10 strategic objectives across six implementation domains. The NpUI emphasizes coordination, education, research excellence, and ethical deployment, aligning closely with the EU AI Act and the OECD AI Principles. Unlike purely reactive compliance models, Slovenia's approach is proactive and visionary. The NpUI embeds AI within the broader goals of the Digital Slovenia 2020 strategy and the Slovenia 2030 Development Strategy, treating AI not just as a digital policy issue but as a societal transformation tool. This reflects a



distinctive national model that mirrors Spain's structured institutional engagement and ethical framing, while emphasizing Slovenia's tradition in AI research dating back over 40 years. A defining trait of Slovenia's model is its emphasis on ecosystem-building. The NpUI aims to create a dynamic AI innovation environment by connecting academic institutions, private sector R&D, civil society, and public authorities. This ecosystem approach seeks to ensure AI deployment is trustworthy, economically productive, and socially beneficial. Slovenia also adopts a context-sensitive, risk-based orientation toward AI governance. The NpUI outlines concrete initiatives in priority areas such as healthcare, agriculture, education, the environment, and smart public services. These efforts are designed not only to drive innovation but also to prepare for the socio-economic disruptions AI may cause—such as labor market shifts and data governance risks.

Much like Spain's AESIA, Slovenia is preparing regulatory infrastructure in response to the EU AI Act, which came into force in August 2024. The Ministry of Digital Transformation has been designated the lead authority for its implementation, supported by a newly established expert council. Slovenia is also preparing an AI regulatory sandbox by 2026, intended to test high-risk systems in a safe and adaptable regulatory setting—akin to the experimental frameworks found in Spain and Singapore. Slovenia's AI governance vision is also closely tied to international collaboration and standard-setting. The country hosts the International Research Centre for Artificial Intelligence (IRCAI) under UNESCO, positioning itself as a key contributor to global efforts on sustainable and ethical AI. Participation in the European AI Factories network via SLAIF further reinforces Slovenia's European and global integration.

From a social inclusion perspective, Slovenia places significant focus on capacity building and educational equity. The NpUI includes ambitious goals for reskilling the workforce, embedding AI into academic curricula, and preparing future generations for responsible AI engagement. While there is still work to be done on integrating AI literacy into primary and secondary education, progress is evident in university-level programs and policy-level coordination for lifelong learning. Finally, the NpUI is designed as a living strategy, subject to dynamic revision based on technological developments and societal needs. This adaptable approach ensures responsiveness to evolving ethical, regulatory, and economic challenges—much like Spain's iterative policy design.

Approach to AI Regulation

Slovenia is proactively developing a comprehensive AI governance framework grounded in national strategy and alignment with European Union regulations. As a result of combining institutional preparedness, sectoral engagement, and international cooperation, Slovenia aims to lead in the responsible and innovative use of artificial intelligence technologies.

1. Regulatory and Strategic Frameworks

Slovenia's AI governance approach is centered on the National Programme for the Promotion of the Development and Use of Artificial Intelligence by 2025 (NpUI), adopted in May 2021. This strategy outlines ten strategic objectives across six implementation areas, ranging from research and innovation to international collaboration. It establishes the foundational structure for national AI development while closely aligning with the European Artificial Intelligence Strategy and the EU Coordinated Plan on AI. To ensure alignment with the EU AI Act, which came into force in August 2024 and will be fully applicable by 2026, Slovenia has drafted a national act to transpose the regulation into domestic law. A public consultation was held in early 2025 to inform this legislative process. Though the NpUI is not formally categorized as a structural reform, it plays a critical role in guiding Slovenia's strategic direction in AI policy.

2. Institutional Support and Strategic Missions

Slovenia has designated the Ministry of Digital Transformation (MDT) as the lead authority for the implementation of the EU AI Act. An expert council has also been established to advise on compliance and ensure stakeholder coordination. The country is preparing the launch of a regulatory sandbox by 2026 to enable secure experimentation with AI technologies and foster innovation within a controlled legal environment. These institutional efforts are supported by inter-ministerial coordination and active involvement of public research organizations and private R&D institutions, creating a cross-sectoral governance model that reflects a whole-of-government approach.

3. Standards Development and Global Engagement

Slovenia contributes to AI standards and governance efforts through its participation in EU-level initiatives and its hosting of the International Research Centre for Artificial Intelligence (IRCAI) under the auspices of UNESCO. Through this, Slovenia plays a prominent role in shaping global AI norms, especially in the context of sustainable development and ethical AI use. The country is also actively involved in the European AI Factories network, with the Slovenian AI Factory (SLAIF) serving as a national innovation hub to support AI deployment in line with EU best practices.

4. Sectoral and Extended Regulatory Approaches

Slovenia's NpUI identifies priority sectors for AI integration, including agriculture, digital economy, corporate governance, education, health, and the environment. Within these sectors, AI is expected to drive innovation and efficiency. Efforts to integrate AI into public administration are ongoing, supported by initiatives such as open access to public sector data and awareness campaigns on AI risks and benefits. Although AI-specific provisions have not yet been integrated into Slovenian data protection, consumer rights, or labor laws, the broader alignment with the EU AI Act ensures these domains will be increasingly addressed as implementation progresses.

5. Fostering Innovation and Indigenous Development

Slovenia's approach to fostering AI innovation includes the planned development of an AI regulatory sandbox and the establishment of the Digital Innovation Hub. These initiatives aim to provide a platform for safe testing of high-risk AI systems and accelerate responsible innovation. With a projected annual AI budget of



€100–500 million, the NpUI emphasizes the importance of private sector funding and public-private partnerships in driving sustainable growth. The government's strategy also supports the development of technical infrastructure, including AI reference solutions and smart data ecosystems to reinforce Slovenia's international competitiveness.

6. AI in Public Sector Transformation

AI integration into the public sector is a central feature of Slovenia's strategy. Key actions include the digitization of public services, AI training for public servants, and improved data access to support transparency and innovation. These reforms position the Slovenian public administration as a proactive actor in AI adoption. Awareness campaigns have also been launched to build public trust and understanding of AI, further reinforcing a responsible and human-centric approach to public sector transformation.

7. AI Skills, Education and Inclusion

Human capital development is a cornerstone of the NpUI. Slovenia invests in AI-focused education, training, and reskilling to prepare its workforce for an AI-driven economy. These efforts are informed by OECD AI principles, promoting inclusive growth, sustainable development, and human well-being. Slovenia's emphasis on education ensures that digital literacy and AI competencies become foundational elements of its long-term innovation policy, with future initiatives expected to deepen these commitments.

8. Infrastructure, Compute and Data Foundations

Slovenia is investing in digital infrastructure to support AI deployment, including the development of a technical framework for reference solutions and national data platforms. These efforts are part of a broader vision to create a scalable and secure AI ecosystem aligned with EU requirements for trusted data spaces and interoperable systems. The upcoming AI sandbox and digital innovation infrastructure will provide essential tools for testing, compliance, and responsible AI scaling, further reinforcing Slovenia's role in the European AI landscape.

Approach to Foster Innovation

Slovenia's comprehensive approach to AI governance, deployment, and innovation is rooted in EU alignment, ethical safeguards, and institutional coordination. The National Programme for the Promotion of the Development and Use of Artificial Intelligence by 2025 (NpUI) positions AI as a driver of social well-being, economic competitiveness, and public sector modernization, while centering trust, human-centricity, and international cooperation in its implementation.

1. Unlocking Economic Value through Targeted AI Deployment

Slovenia's NpUI integrates AI across high-potential sectors such as healthcare, public administration, agriculture, environment, mobility, and education. These sectors were selected based on their capacity to deliver strong societal and economic returns. The programme emphasizes "reference AI solutions" that can be scaled and adapted nationally, particularly within the public sector. AI is framed not only as a productivity enhancer but as a public good capable of transforming service delivery, driving innovation, and improving citizens' quality of life.

2. Catalysing Innovation through Strategic Clustering and Ecosystems

The NpUI fosters a national AI ecosystem by linking universities, public research institutes, industrial R&D centers, and digital innovation hubs (DIHs). It actively supports collaboration between these actors through Strategic Research and Innovation Partnerships (SRIPs), notably in domains like Smart Cities and Factories of the Future. These clusters act as platforms for co-creation, experimentation, and knowledge transfer—connecting Slovenia's research legacy with economic deployment, while reinforcing regional cohesion and sectoral competitiveness.

3. Enabling Trusted and Responsible Data Use

Slovenia's AI programme embeds robust data governance, transparency, and accountability. It advocates open access to public sector data, interoperability of systems, and alignment with GDPR and EU digital policies. The development of secure, FAIR (Findable, Accessible, Interoperable, Reusable) data environments is prioritized. AI applications must comply with ethical and legal standards, supporting public trust in data-driven innovation and aligning with Slovenia's broader digital rights commitments.

4. Scaling AI in the Public Sector

Public administration is both a priority domain and a demonstrator for responsible AI. The NpUI promotes the introduction of AI in service design, policy planning, and citizen engagement, targeting improvements in efficiency, responsiveness, and quality. Training programs for public officials and awareness campaigns for citizens support a digitally mature public sector. Slovenia's state-led use of AI in domains like language technologies, healthcare, and public registries reflects its ambition to model trustworthy AI adoption.

5. Advancing Compute and Research Infrastructure

Slovenia's investments in compute infrastructure include the HPC RIVR supercomputing project and participation in the EuroHPC initiative. These resources support the development and testing of AI solutions at scale. The NpUI also supports national data platforms and standardization efforts, fostering alignment with EU-wide infrastructures. Slovenia's strong academic foundation, including over 40 years of AI research at leading institutions like Jožef Stefan Institute and University of Ljubljana, underpins the country's commitment to scientific excellence and technological sovereignty.

6. Empowering SMEs and Startups

AI is positioned as a key enabler for the digital transformation of Slovenia's SME sector. The NpUI promotes pre-commercial procurement, funding for AI solution development, and public-private partnerships that empower small firms to adopt AI responsibly. By embedding SMEs in national innovation networks and offering access to technical infrastructure and expertise, Slovenia supports business model transformation and international competitiveness for its innovation-driven enterprises.

7. Developing a Future-Ready Workforce

Workforce development is central to the NpUI, with a strong focus on AI education and digital literacy. The strategy supports university-level AI programs and lifelong learning initiatives, while also calling for greater integration of AI-related topics in primary and secondary education. Coordination across ministries ensures that emerging skills gaps are addressed. Gender equality and inclusion are acknowledged, though further work remains to structurally embed these across educational pipelines.

8. Aligning AI Innovation with Sustainability Goals

Slovenia's AI strategy intersects with national and EU climate goals by promoting AI solutions in environmental monitoring, smart agriculture, green mobility, and disaster resilience. Sustainability is embedded in both technological and policy dimensions of the NpUI. The programme supports the development of energy-efficient AI systems and recognizes AI's potential to address challenges of the green transition, in line with the European Green Deal.

9. Building Public Trust and Ethical AI

Trust and human-centricity form the ethical backbone of Slovenia's AI programme. The NpUI promotes ethical AI design through legal frameworks, transparency guidelines, and public engagement. The programme aligns with OECD and UNESCO principles and calls for continuous societal dialogue on the values shaping AI use. Ethical oversight and citizen awareness efforts are integrated across sectors, aiming to ensure that technological advancement reinforces—not undermines—democratic values and human rights.

Digital Decade

2025: Report

AI remains a top priority for Slovenia within the broader field of advanced technologies. With its successful application to host an AI Factory and the initiation of the SMASH project (Machine Learning for Science and Humanities), Slovenia is well-positioned to further expand its contributions to AI development. The AI Factory is set to offer AI tools and computing capacity to support both businesses and public administration. Building on the existing capabilities of the Vega supercomputer, a new infrastructure is being developed to enhance research and industrial applications of AI—such as climate forecasting and accelerated process simulation in sectors like manufacturing.

Momentum is also building around other AI-related initiatives. Following an extensive planning phase starting in 2023, Slovenia will publish its first tender call in 2025 for the AI Competence Centre, with a budget of EUR 3.6 million. This Centre will support a wide range of activities including AI training, business adoption, ecosystem development, and international cooperation. Aligned with the Digital Decade targets, it aims to help companies integrate and advance their AI usage by offering training programmes, consulting services, and practical AI tool guidance. Stakeholders appear to be in agreement on the Centre's scope and

objectives, and more tenders are expected in the coming years. Public sentiment strongly supports these efforts: according to the 2025 Eurobarometer, 80% of Slovenians believe public authorities should actively steer the development of AI and other digital technologies to ensure alignment with fundamental rights and values.

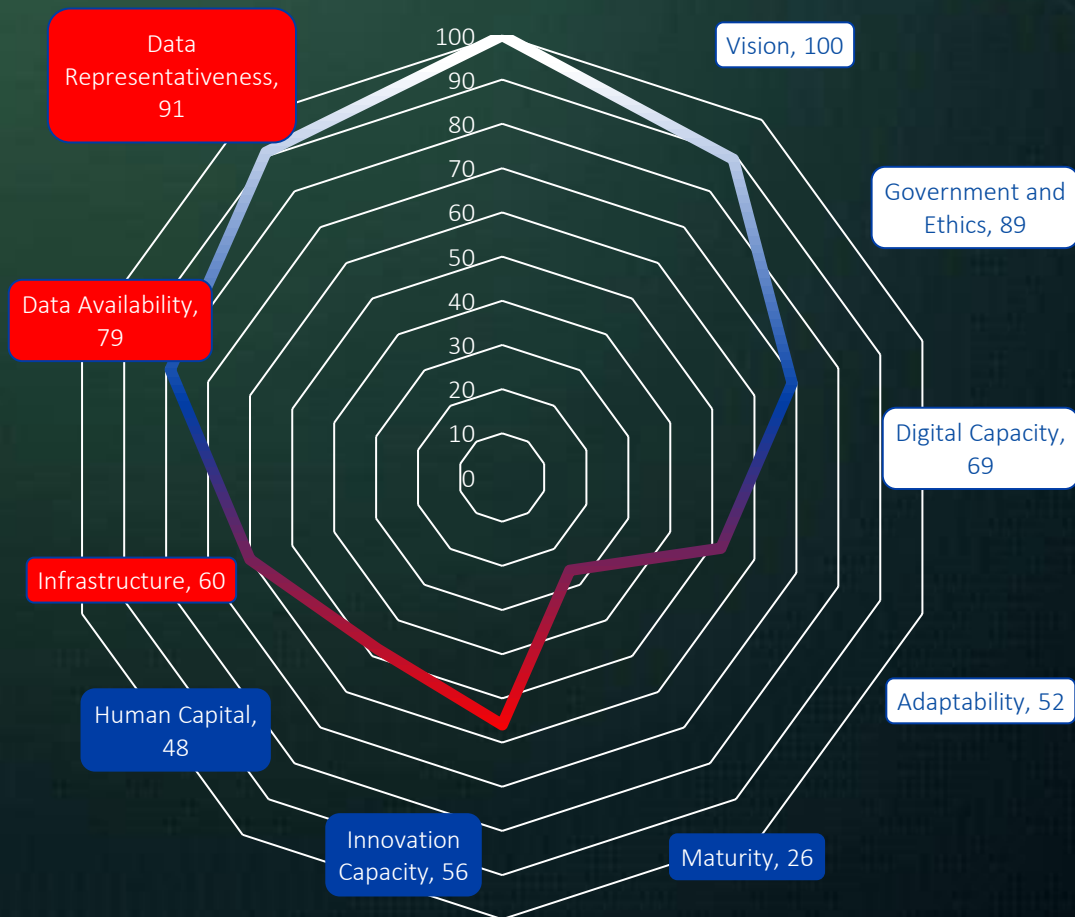
In its updated roadmap, Slovenia has introduced a new measure titled ‘Support for low-value AI applied projects’, replacing a previously planned initiative on vocational training. Under this new scheme, EUR 6.4 million will be allocated to help around 70 companies develop AI-based solutions—from early-stage testing to commercial deployment. The goal is to broaden the availability of AI technologies and enhance internal capabilities within those companies, contributing to increased AI uptake across the national economy. The EU recommended that Slovenia accelerate efforts to foster adoption of advanced technologies, particularly by enhancing actions in data analytics and by expediting and refining AI-related measures. Slovenia responded to this guidance with several new policy actions in 2024, focusing heavily on AI. The country continues to implement projects to boost advanced technology adoption, with further activities included in the revised roadmap.



Government AI Readiness

‘Eastern Europe ranks 4th in this year’s Government AI Readiness Index, with an average score of 57.88—over 10 points above the global average (47.59). Estonia (72.62) leads the region and is the only Eastern European country to feature in the global top 25. The region performs above the global average across all three pillars. Its strongest performance is in the Data & Infrastructure pillar (72.10), making it the third-best performing region globally in this area. Eastern Europe also shows solid results in the Government pillar (62.52), where it again ranks third worldwide. However, the Technology Sector remains a challenge, with a regional score of 39.03—more than 20 points behind the other two pillars—and ranking fifth globally, falling narrowly behind the Middle East and North Africa (39.34). This reflects the need for further investment in technological capacity and innovation to unlock the region’s full AI readiness potential.

2024 Scores by Dimension



Government

Slovenia scores 77.48 in the Government pillar, positioning it among the top performers in Eastern Europe and globally. This reflects its strong policy coordination, active participation in EU-level governance, and clear national commitment through the National Programme for the Promotion of the Development and Use of Artificial Intelligence by 2025 (NpUI). Adopted in 2021, the NpUI sets out 10 strategic objectives covering education, ethical AI, international cooperation, and public sector digital transformation. Slovenia is also aligning its legal framework with the EU AI Act, designating the Ministry of Digital Transformation as the lead authority and establishing an expert council to oversee its implementation. The planned AI regulatory sandbox and the hosting of the UNESCO-affiliated IRCAI underline Slovenia's role as a proactive and values-driven contributor to European AI governance.

Technology Sector

Slovenia's score of 43.42 in the Technology pillar indicates a developing innovation ecosystem with long-standing academic strengths but relatively modest commercialization and scaling of AI technologies. While Slovenia has over 40 years of AI research excellence through institutions like the Jožef Stefan Institute and the University of Ljubljana, translating this into industrial applications remains a challenge. The NpUI identifies gaps in investment, startup support, and cross-sectoral deployment, and calls for strengthened public-private partnerships, support for SMEs, and integration of AI in key sectors such as manufacturing, healthcare, and logistics. The country is also working to deepen its participation in EU research programmes and innovation networks such as SLAIF to expand its technological footprint.

Data & Infrastructure Sector

Scoring 76.76, Slovenia performs exceptionally well in the Data & Infrastructure pillar. It boasts robust digital infrastructure, including its participation in the EuroHPC initiative and national investments like the HPC RIVR supercomputing project. The NpUI prioritizes the development of interoperable systems, FAIR data governance practices, and open public data platforms. Slovenia's digital strategy includes broadband expansion, smart public services, and cloud-based infrastructure for secure and scalable AI deployment. The country's commitment to international standards and participation in data-driven initiatives at both the EU and OECD levels further enhance its readiness in this domain.

With a total score of 65.85, Slovenia ranks above both the Eastern European (57.88) and global (47.59) averages, making it one of the regional leaders in AI governance and readiness. Its strengths lie in institutional coordination, data infrastructure, and a forward-looking national strategy that aligns tightly with EU regulations and ethical frameworks. While technology-sector scale-up remains an area for improvement, Slovenia's foundation in research, regulatory preparedness, and public sector innovation positions it well for accelerated AI integration. Future success will depend on increased private-sector engagement, targeted investments in AI capabilities, and continued alignment with EU digital policy objectives.

Conclusion

Slovenia's artificial intelligence ecosystem is evolving with precision—anchored in ethical governance and European harmonisation, while gradually accelerating its innovation trajectory. Grounded in a values-driven strategy, Slovenia has adopted the National Programme for the Promotion of the Development and Use of Artificial Intelligence by 2025 (NpUI), which sets out ten strategic objectives across sectors and lays the groundwork for a comprehensive, rights-based AI framework. With the Ministry of Digital Transformation designated as the lead authority for the EU AI Act and the establishment of an expert council, Slovenia has built a solid institutional core for responsible AI oversight.

Slovenia ranks among the top Eastern European countries in data availability and digital infrastructure, thanks to strategic investments in national projects like HPC RIVR and participation in initiatives such as EuroHPC and the European AI Factories network. These developments underpin scalable AI deployment in priority sectors including public administration, healthcare, environment, and education—domains where AI is already enhancing service design, sustainability, and citizen engagement. Slovenia's governance approach—deeply rooted in



transparency, human dignity, and public trust—demonstrates a national commitment to embedding democratic values into the AI lifecycle.

Yet key challenges persist. Slovenia's modest technology sector score reflects the ongoing difficulty in translating its 40-year legacy of AI research excellence into commercial applications and global competitiveness. Despite strong academic foundations, industrial integration and startup momentum remain limited. To realise its full AI potential, Slovenia must strengthen SME involvement, enhance cross-sector innovation clusters, and scale public-private partnerships. Sustained investment in compute capacity, AI reference solutions, and smart data ecosystems will be critical for catalysing growth and translating national strategies into tangible economic outcomes.

To maintain its forward-looking trajectory, Slovenia must continue deepening AI literacy, expanding lifelong learning, and embedding inclusivity and gender equity in workforce development. Regional coordination across ministries and sectors will also be vital in ensuring that innovation benefits the whole of society, not just urban or institutional hubs. Internationally, Slovenia's leadership in AI standard-setting—through its hosting of the UNESCO-affiliated IRCAI and active engagement in EU and OECD frameworks—positions the country as a bridge between ethical governance and global cooperation.

If backed by expanded private-sector participation, and agile implementation of the EU AI Act, Slovenia can cement its role as a regional model for trustworthy AI.

Appendix A:

Expert Vignettes

This appendix presents short expert vignettes that illustrate how Slovenia's institutional landscape for AI oversight, coordination and policy implementation operates in practice. Each vignette is written from the perspective of a key AI ecosystem typical actor) and highlights institutional responsibilities, accountability arrangements and multi-stakeholder mechanisms that shape trustworthy, inclusive and adaptive AI governance



Andrej P. Skraba, Co-Founder, Astra AI

Slovenia has the opportunity to lead in AI for education: blending world-class research, agile startups, and a tradition of learning, we can pioneer personalized, scalable solutions shaping how future generations everywhere learn and thrive.



Biography

Andrej P. Skraba is the co-founder of Astra AI, an edtech startup harnessing artificial intelligence to revolutionize how people learn. A seasoned CEO, he has spent over a decade in the high-tech industry. Today, his passion is building AI-powered educational solutions that make knowledge more accessible, personalized, and engaging. At Astra AI, Skraba's leadership philosophy centers on fostering product-oriented teams, guided by analytics, KPIs, and continuous user feedback. He champions a culture of experimentation, adaptation, and growth, ensuring innovation is always tied to meaningful learning outcomes. By blending technological expertise with a people-first mindset, Skraba is dedicated to shaping the future of education and empowering learners worldwide..

Kristina Kočet Hudrap, CEO, TIKO PRO

Artificial intelligence is no longer the future — it's our present, fundamentally reshaping how we work, make decisions, learn, and even design public policies. Slovenia has the knowledge, a strong research base, and agile companies, but the full potential of AI is still untapped. In the next 1–2 years, we need to focus on accelerating AI adoption in business, investing in digital skills, and building clearer, more flexible regulatory support.



Biography

Kristina Kočet Hudrap, is a prominent entrepreneur and leader at Tiko Pro d.o.o., specializing in EU funding acquisition. With over 92 years of combined experience in her team, she has successfully guided numerous clients in Slovenia, Croatia, and Austria to secure funding for over 800 projects, achieving a high success rate. Her social media presence emphasizes leadership, sustainability, and innovation, often sharing insights on entrepreneurship and the importance of empowering the next generation of business leaders.

Sara Kremsar, *Founder,* Optimod

AI progress is measured by building systems that last, not by how fast we move. Slovenia already has the knowledge, research foundations, and values to work with AI responsibly. The real work now is connecting these strengths across education, industry, and public institutions to create shared, long-term impact that benefits society as a whole.



Biography

Sara Kremsar, M.Sc., Sara Kremsar, M.Sc., is the founder of Optimod, a company that helps fast-growing organizations build clarity, structure, and alignment as they scale. Her work focuses on organizational structures, business processes, and system optimization, supporting companies across various sectors since 2016 to improve efficiency, strengthen leadership, and enable sustainable growth. She founded Optimod in 2017 to connect research, education, and real-world entrepreneurship. Through initiatives such as the Improving Entrepreneurial Journey Initiative (IEJI), Optimod fosters the exchange of practical experience and academic insight. Her vision is to build resilient, human-centered ecosystems where collaboration, learning, and innovation help individuals and organizations reach their full potential.

Appendix B: OECD AI Progress Report

This note summarises the OECD country note on **Slovenia's** progress implementing its 2021–26 national AI strategy (the “National Programme on Artificial Intelligence, NpAI”) and how those commitments are being integrated into broader national data, research and digital skills initiatives.

Slovenia's national AI strategy, the *NpAI* (adopted in 2021), envisions a systemic, horizontal approach rather than a purely sectoral one. It allocates **€112 million** in investments (including ~€65 million from the EU Recovery & Resilience Facility) to support research, infrastructure, trustworthy AI and public-sector adoption.

The country is renewing its strategy beyond 2026, extending to 2030, with an updated implementation plan that aligns with its **Digital Slovenia 2030** strategy and the National Strategic Plan for the Digital Decade.

On the research front, Slovenia benefits from strong institutions, including the Jožef Stefan Institute, several universities, and a planned **AI Competence Centre**. The Competence Centre (due 2024–29) aims to serve as a “one-stop shop” for both public-sector and private-sector adoption of AI.

In infrastructure, Slovenia is building up **high-performance computing (HPC)** capacity, most notably via the **Vega** supercomputer at IZUM, under the EuroHPC Joint Undertaking, to support machine learning, deep learning, and big-data analytics.

On skills and education, Slovenia is integrating AI across its education system:

- Primary and secondary curricula are being reformed to include computing, digital skills, and “AI content” in various subjects.
- There is a major **requalification programme for women in ICT**, launched in 2023, which includes AI training.
- At the higher-education level, reforms are underway to support micro-credentials, enabling more flexible retraining/upskilling in AI-related disciplines.

AI adoption is already occurring in several high-impact sectors. Notably:

- **Healthcare:** Slovenia has a mature health-data infrastructure (via its National Healthcare Database), supports cross-border health data flows, and is building secure data environments aligned with European Health Data Space principles.
- **Public sector:** A modular, horizontal AI platform (“semantic analyser”) is being piloted to support public administration via natural-language processing.
- **Energy and environment:** Pilot projects use AI for energy efficiency, renewables, and decarbonisation.
- **Agriculture / Bioeconomy:** For example, the *Green.Dat.AI* Horizon Europe project supports digital twins for smart farming and water management.
- **Mobility:** Traffic-monitoring AI projects are underway (e.g., for congestion prediction and accident prevention).

Finally, governance: the NpAI sets out ten strategic goals, including establishing a national AI observatory, strengthening public trust, developing an ethical and legal framework, and ensuring international co-operation.

Strengths

- **Comprehensive, systemic vision:** Rather than focusing on a few sectors, Slovenia’s strategy takes a horizontal approach, touching on infrastructure, research, public trust, ethics, and education.
- **Strong research base:** Long-standing AI research capacity (Jožef Stefan Institute, universities) and the forthcoming AI Competence Centre provide solid institutional foundations.

- **Growing HPC infrastructure:** Building the Vega supercomputer demonstrates concrete commitment to compute capacity.
- **Education & skills pipeline:** Integration of AI into school curricula, support for retraining (especially women), and micro-credentials in higher education show a multi-layered approach to skills.
- **Sectoral leadership:** Proactive AI adoption in healthcare, energy, mobility, agriculture illustrates real-world deployment.
- **Alignment with broader strategies:** The NpAI is well integrated into Slovenia's **Digital Slovenia 2030** strategy and the EU's digital decade plan.

Constraints and Risks

- **Strategy–implementation gap:** While the NpAI is detailed, scaling from high-level commitments to concrete, deliverable projects (especially post-2026) could be challenging.
- **HPC consolidation risk:** Building Vega and scaling the HPC infrastructure is capital-intensive; under-utilisation or slow adoption by local SMEs or public institutions could limit the return.
- **Public trust and ethical governance:** Although the strategy addresses ethics, deploying AI in sensitive domains like healthcare or public administration requires continuous, transparent governance and mechanisms for public engagement.
- **Talent bottlenecks:** Despite initiatives, demand for AI-specialists, data engineers and HPC-savvy researchers may outstrip supply, risking brain drain or unfilled roles.
- **SME adoption obstacles:** Start-ups and SMEs may lack the capacity (financial, human, technical) to take full advantage of AI funding or the AI Competence Centre, especially without targeted support.
- **Cross-sector coordination:** The horizontal design of the strategy is a strength, but also requires robust inter-ministerial coordination; failure here could fragment efforts.

Practical Recommendations (Next Steps)

1. Develop a detailed 2026–2030 roadmap

- Publish a time-bound implementation plan with clear milestones, assigned responsibilities (which ministry / agency leads each measure), and budget allocations.
- Use the interministerial working group to monitor progress and produce annual public reports on progress.

2. Accelerate HPC access and utilisation

- Ensure Vega's compute capacity is accessible not just to research institutions but to SMEs and the public sector.
- Establish fellowship programmes (e.g., attract international AI researchers, returning Slovenians) to use HPC for AI projects.

3. Operationalise the AI Competence Centre

- Ensure the centre becomes a “one-stop shop” for adoption: advisory services, pilot funding, matchmaking between research and industry.
- Encourage public–private partnerships via the Competence Centre to drive real-world deployments.

4. Enhance data governance and trust mechanisms

- Complement the strategy with a governance framework that codifies ethical AI, accountability, transparency, and citizen engagement.
- Consider establishing an **AI observatory**, as foreseen, that publishes impact assessments, risk registers, and sectoral benchmarks.

5. Scale educational and upskilling programmes

- Expand micro-credential offers in AI / data science in universities and vocational training.
- Increase the requalification programme for underrepresented groups (e.g., more funding, broader outreach to women, rural populations).

- Embed AI literacy in school curricula more systematically, and support teachers via training and open educational resources.

6. Boost sectoral adoption via pilot-to-scale models

- Use the Competence Centre to run pilot projects in public sector, healthcare, energy, mobility; then document, evaluate, and scale successful pilots.
- Provide matched funding or public procurement incentives for SMEs to adopt AI solutions developed in pilot contexts.

Overall Assessment

According to the OECD, **Slovenia has made strong progress** in building the foundations for a national AI ecosystem: it has committed substantial funding, is investing in HPC infrastructure, has credible research institutions, and is integrating AI into its education system.

Its horizontal, systemic strategy (rather than narrow sectoral focus) gives it flexibility and coherence, but the challenge now is **moving from strategy to execution**: establishing a clear roadmap for the post-2026 phase, operationalising its AI Competence Centre, ensuring widespread and equitable access to compute, deploying in critical sectors, and embedding governance and trust mechanisms.

If Slovenia can sustain momentum — by translating strategic ambitions into measurable projects, scaling adoption, and maintaining public accountability — it is well-positioned to leverage AI for sustainable economic and social transformation.



References

This section compiles all primary and secondary sources cited—legal texts, policy papers, academic studies, and official reports. It provides a transparent basis for the report’s analyses and recommendations, enabling verification, further research, and continued dialogue on AI policy and governance in Slovenia.

Center for AI and Digital Policy, (2025), ‘Artificial Intelligence and Democratic Values 2025’, accessible at: <https://www.caidp.org/reports/aidv-2025/> (last accessed 3rd May 2025)

Digital Skills & Jobs Platform, (2025), ‘Slovenia - PURITY: National programme to promote the development and use of AI by 2025’, accessible at: <https://digital-skills-jobs.europa.eu/en/actions/national-initiatives/national-strategies/slovenia-purity-national-programme-promote> (last accessed 23rd December 2025)

European Commission, (2025), ‘Digital Decade 2025: Country reports’, accessible at <https://digital-strategy.ec.europa.eu/en/library/digital-decade-2025-country-reports> (last accessed 9th June 2025)

European Parliament and The Council of the European Union, (2024), 2024/1689 Regulation (EU) 2024/1689 of the European Parliament and of The Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act), accessible at https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_202401689 (last accessed 9th June 2025)

OECD, (2025), 'Progress in Implementing the European Union Coordinated Plan on Artificial Intelligence (Volume 1): Slovenia', accessible at: https://www.oecd.org/en/publications/progress-in-implementing-the-european-union-coordinated-plan-on-artificial-intelligence-volume-1_6d530a88-en/slovenia_9f609de3-en.html (last accessed 20th November 2025)

OECD.AI, (2025), 'AI in Slovenia', accessible at: <https://oecd.ai/en/dashboards/countries/Slovenia> (last accessed 9th June 2025)

Oxford Insights, (2024), 'Government AI Readiness Index', accessible at: <https://oxfordinsights.com/ai-readiness/ai-readiness-index/> (last accessed 9th June 2025)

Republic of Slovenia, (2020). 'National Programme to Promote the Development and Use of Artificial Intelligence in the Republic of Slovenia by 2025 (NpAI)', accessible at: https://www.gov.si/assets/ministrstva/MDP/National_Programme_for_AI_2025.pdf (last accessed 9th June 2025)

Stanford University: Human-Centered Artificial Intelligence (HAI), (2025)) 'AI Index', accessible at: <https://hai.stanford.edu/ai-index> (last accessed 3rd May 2025)



Important notice

This document has been prepared by AI & Partners B.V. for the sole purpose of enabling the parties to whom it is addressed to evaluate the capabilities of AI & Partners B.V. to supply the proposed services.

Other than as stated below, this document and its contents are confidential and prepared solely for your information, and may not be reproduced, redistributed or passed on to any other person in whole or in part. If this document contains details of an arrangement that could result in a tax or National Insurance saving, no such conditions of confidentiality apply to the details of that arrangement (for example, for the purpose of discussion with tax authorities). No other party is entitled to rely on this document for any purpose whatsoever and we accept no liability to any other party who is shown or obtains access to this document.

This document is not an offer and is not intended to be contractually binding. Should this proposal be acceptable to you, and following the conclusion of our internal acceptance procedures, we would be pleased to discuss terms and conditions with you prior to our appointment. Images used throughout the document have either been produced in-house or sourced from publicly available sources (see **References** for details).

AI & Partners B.V. is the Dutch headquarters of AI & Partners, a global professional services firm. Please see <https://www.ai-and-partners.com/> to learn more about us.

© 2025 AI & Partners B.V. All rights reserved.

Designed and produced by AI & Partners B.V.